11/04/2025

Malicious Code Analysis

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CSCI 591

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Overview

01 Debugger Overview

02 WinDbg Preview

Part One

01

Debugger Overview

2025-11-4

Types of Debuggers

- Command line debuggers
  - kd.exe: kernel debugger
  - cdb.exe, ntsd.exe: user mode debugger
- WinDbg
  - GUI on top of kd.exe and cdb.exe
  - Much more efficient for source debugging
  - Can be slower because of extra data displayed

Part Two

02

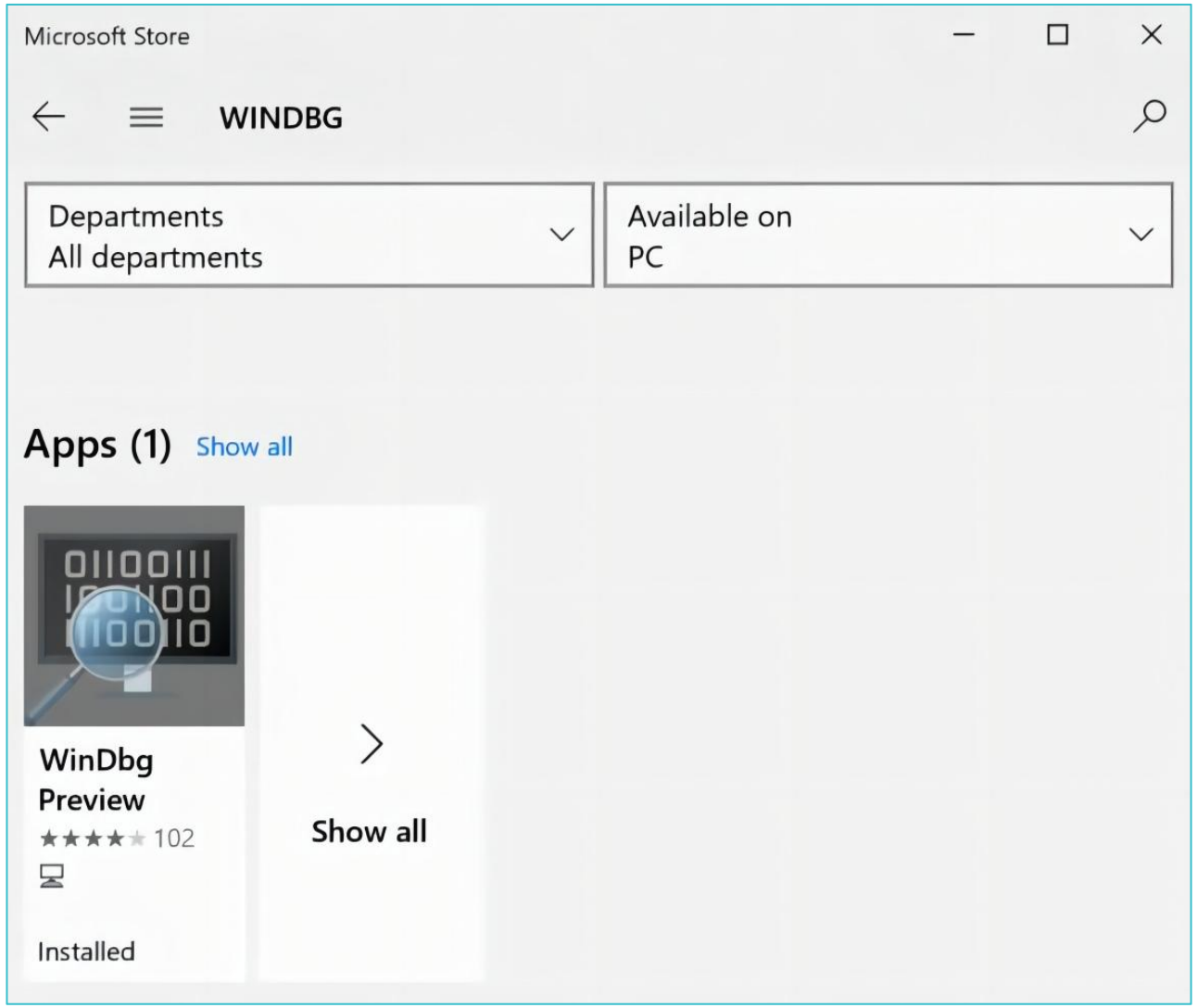
WinDbg Preview

2025-11-4

Installing

- Installing WinDbg Preview
- If you are using the Windows 10 or above, WinDbg Preview is already installed.
- On Windows 10, at the lower left of the desktop, click the magnifing glass. Type STORE
- Open Microsoft Store.
- In Microsoft Store, search for WinDbg, as shown below.

Installing



- When it finds "WinDbg Preview", click the **blue Get** button.
- A "Use across your devices" box pops up. Click "**No, thanks**".
- Click **Launch**.

Launching

- Close WinDbg.
- Click the **Start** button and type **WINDBG**. Right-click "**WinDbg Preview**" and click "**Run as Administrator**".
- Approve the privilege escalation.

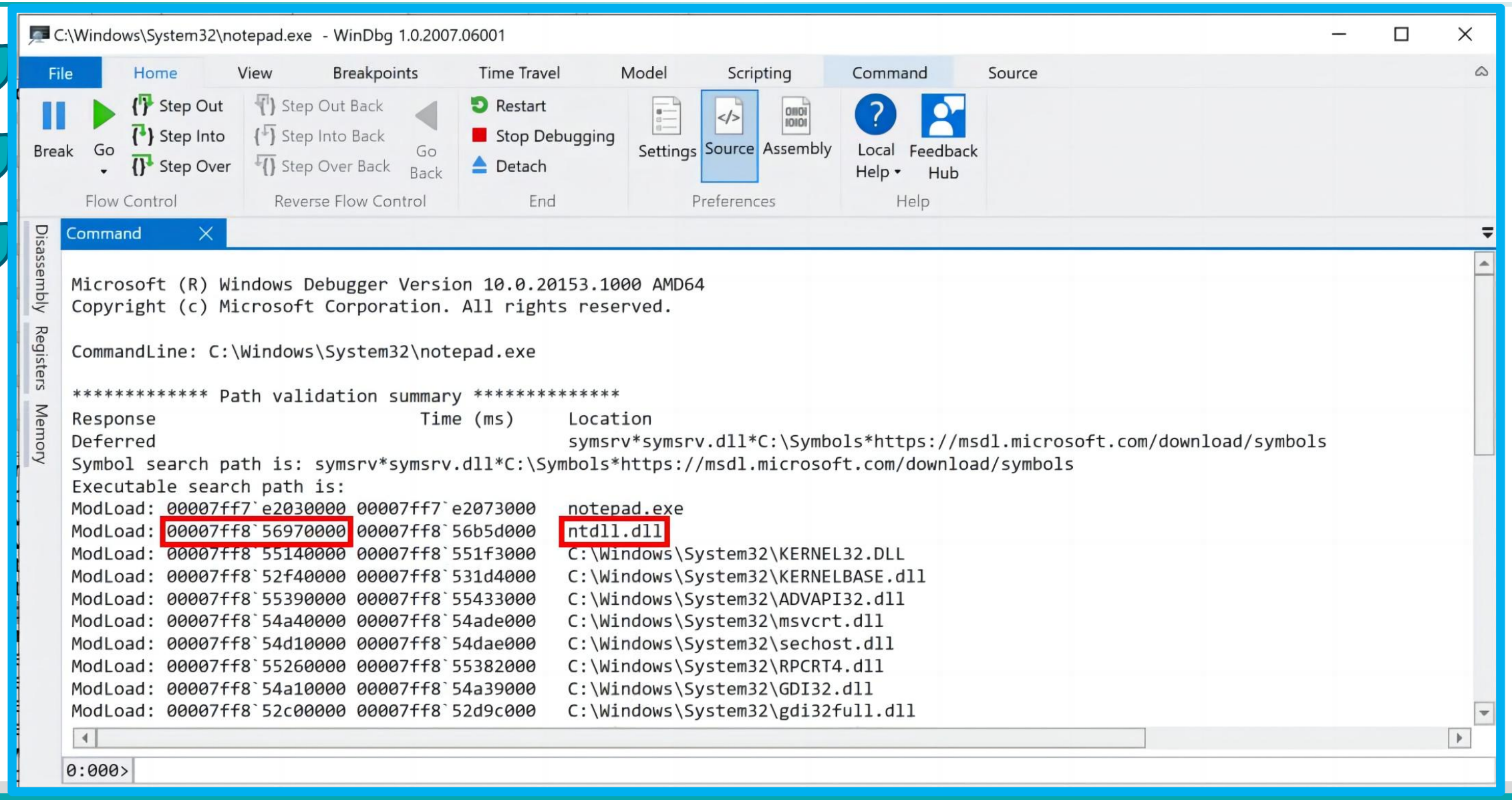
Debugging Notepad

- In WinDbg, click **File, "Launch executable".**
- Navigate to:

  C:\Windows\System32\notepad.exe

  and open it.

Loading DIIs



Loading DlIs

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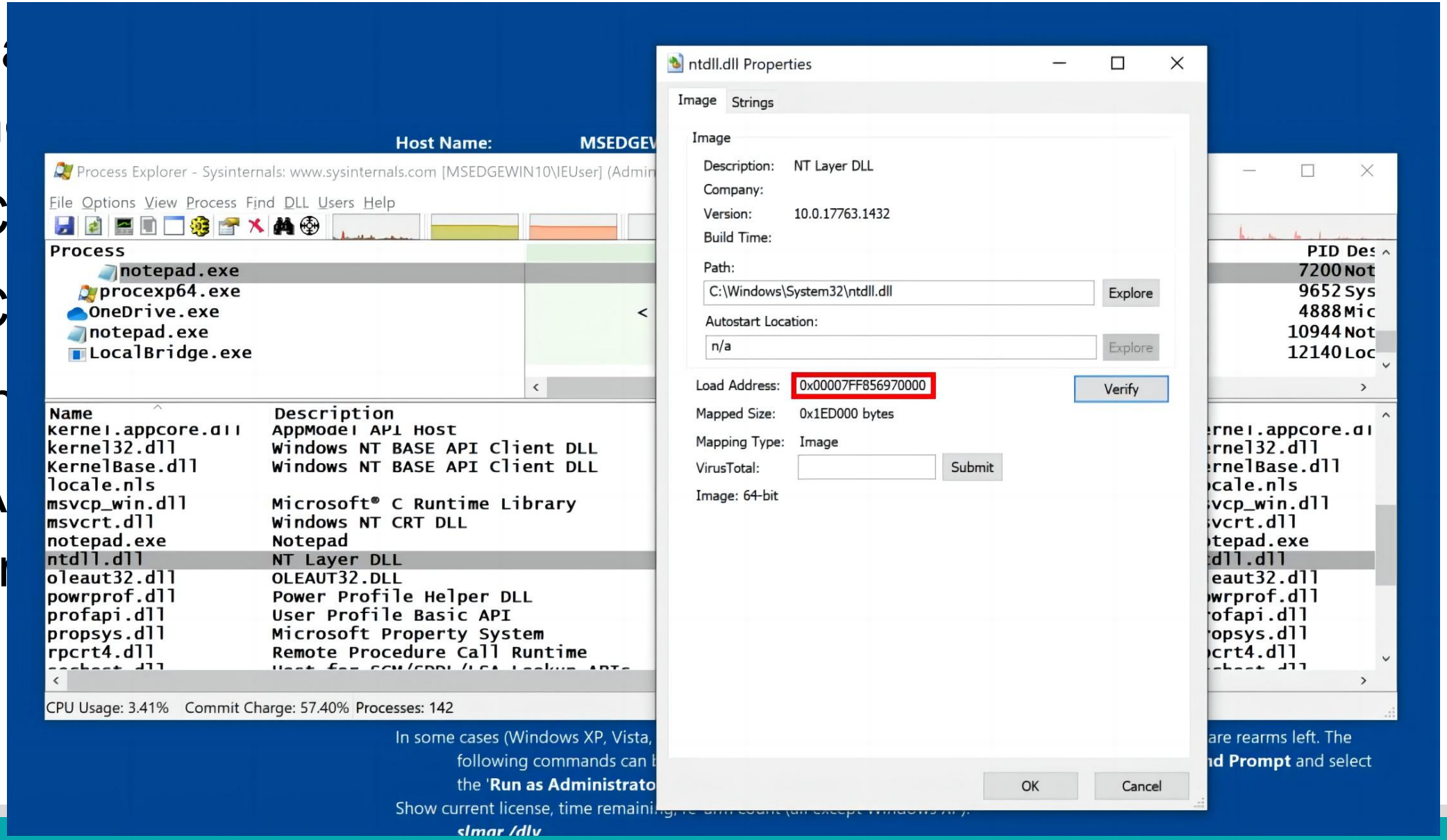

Stack

- In WinDb … he bottom.
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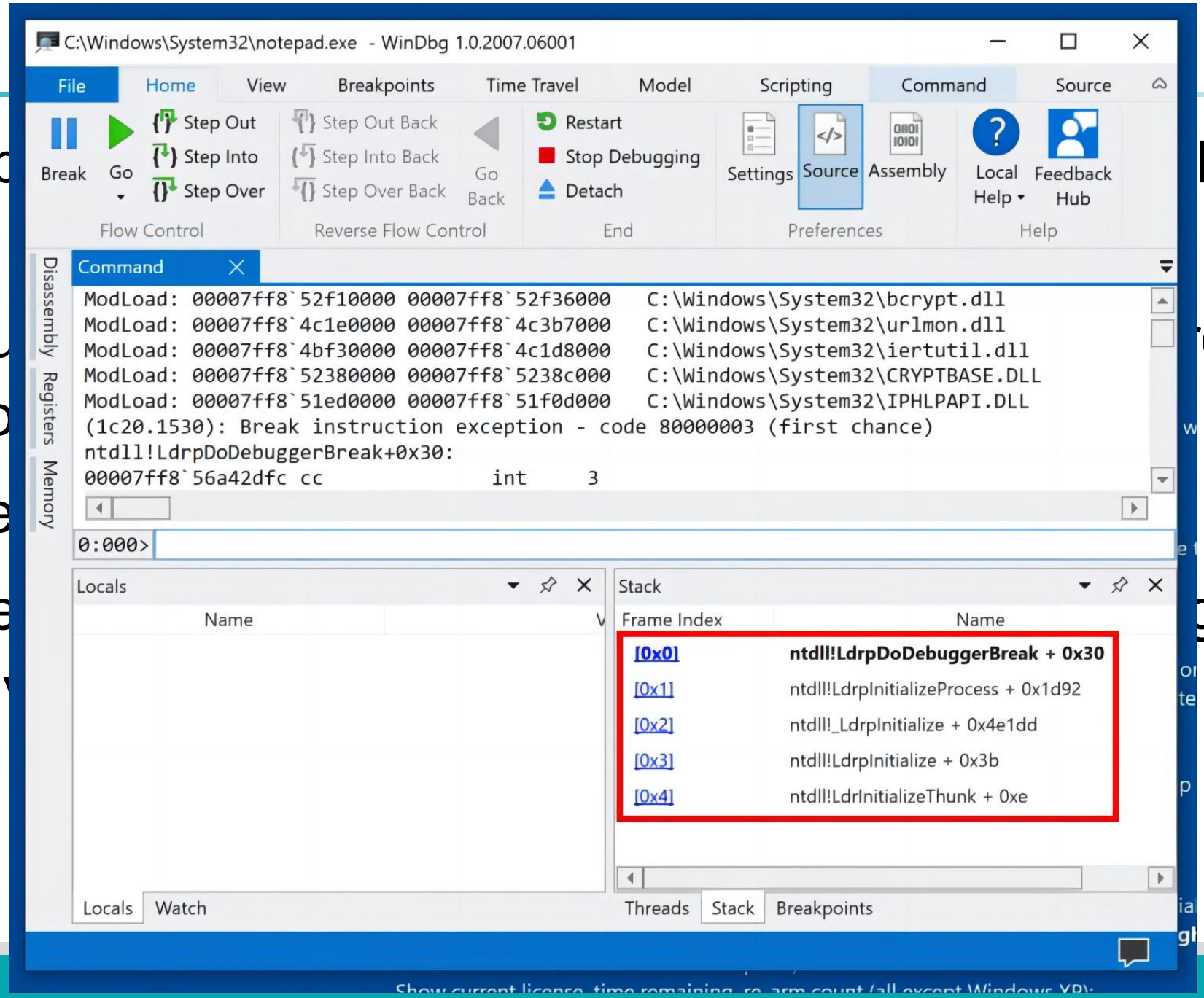
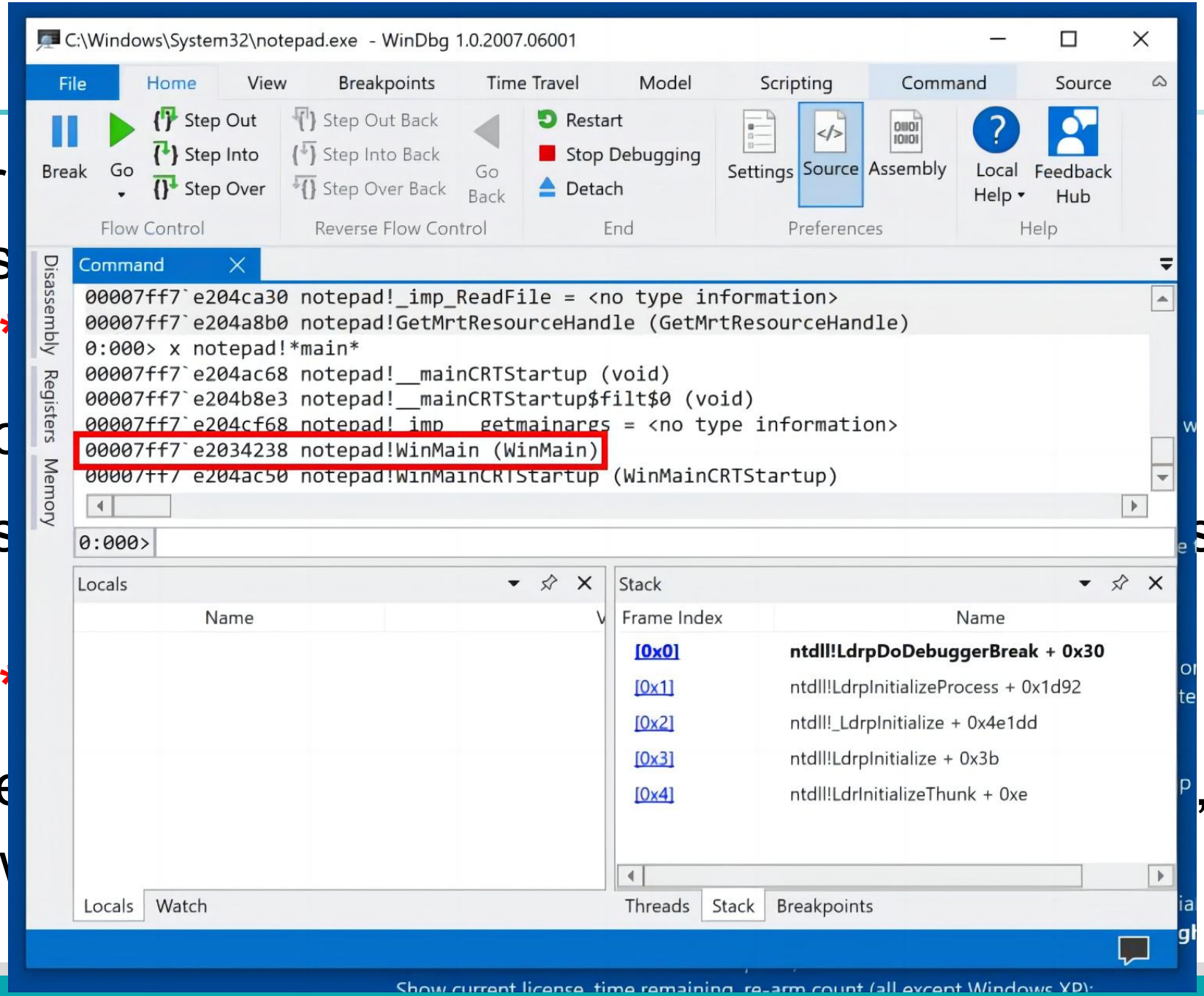


Viewing

- In the lower
  execute this
  x notepad!*
- You see a lo
- To see the s
  command:
  x notepad!*
- You see a fe
  shown below



Setting

- Execute th n, and display bre

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- The break

C:\Windows\System32\notepad.exe - WinDbg 1.0.2007.06001

File | Home | View | Breakpoints | Time Travel | Model | Scripting | Command | Source

Break | Go | Step Out | Step Into | Step Over | Step Out Back | Step Into Back | Step Over Back | Go Back | Restart | Stop Debugging | Detach | Settings | Source | Assembly | Local Help | Feedback Hub

Flow Control | Reverse Flow Control | End | Preferences | Help

Disassembly | Registers | Memory

Command

```
00007ff7`e204b8e3 notepad!__mainCRTStartup$filt$0 (void)
00007ff7`e204cf68 notepad!_imp___getmainargs = <no type information>
00007ff7`e2034238 notepad!WinMain (WinMain)
00007ff7`e204ac50 notepad!WinMainCRTStartup (WinMainCRTStartup)
0:000> bu notepad!WinMain
0:000> bl
     0 e Disable Clear  00007ff7`e2034238     0001 (0001)  0:**** notepad!WinMain
```

0:000>

Locals

Name	V

Stack

Frame Index	Name
[0x0]	**ntdll!LdrpDoDebuggerBreak + 0x30**
[0x1]	ntdll!LdrpInitializeProcess + 0x1d92
[0x2]	ntdll!_LdrpInitialize + 0x4e1dd
[0x3]	ntdll!LdrpInitialize + 0x3b
[0x4]	ntdll!LdrInitializeThunk + 0xe

Locals | Watch | Threads | Stack | Breakpoints

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C:\Windows\System32\notepad.exe - WinDbg 1.0.2007.06001

File | Home | View | Breakpoints | Time Travel | Model | Scripting | Command | Source

Break | Go | Step Out | Step Into | Step Over | Step Out Back | Step Into Back | Step Over Back | Go Back | Restart | Stop Debugging | Detach | Settings | Source | Assembly | Local Help | Feedback Hub

Flow Control | Reverse Flow Control | End | Preferences | Help

Disassembly | Registers | Memory

Command

```
0:000> bl
     0 e Disable Clear  00007ff7`e2034238     0001 (0001)  0:**** notepad!WinMain
0:000> g
ModLoad: 00007ff8`53b90000 00007ff8`53bbe000   C:\Windows\System32\IMM32.DLL
Breakpoint 0 hit
notepad!WinMain:
00007ff7`e2034238 48895c2410      mov     qword ptr [rsp+10h],rbx ss:00000016`bb67f898=0000000000000000
```

0:000>

Locals

Name	Value

Stack

Frame Index	Name
[0x0]	**notepad!WinMain**
[0x1]	notepad!__mainCRTStartup + 0x19f
[0x2]	KERNEL32!BaseThreadInitThunk + 0x14
[0x3]	ntdll!RtlUserThreadStart + 0x21

Locals | Watch | Threads | Stack | Breakpoints

C:\Windows\System32\notepad.exe - WinDbg 1.0.2007.06001

File | Home | View | Breakpoints | Time Travel | Model | Scripting | Command | Source

Break | Go | Step Out | Step Into | Step Over | Step Out Back | Step Into Back | Step Over Back | Go Back | Restart | Stop Debugging | Detach | Settings | Source | Assembly | Local Help | Feedback Hub

Flow Control | Reverse Flow Control | End | Preferences | Help

Disassembly | Registers | Memory

Command

```
00007ff8`54960000 00007ff8`54a09000   shcore     (deferred)
00007ff8`54a10000 00007ff8`54a39000   GDI32      (deferred)
00007ff8`54a40000 00007ff8`54ade000   msvcrt     (deferred)
00007ff8`54be0000 00007ff8`54d07000   COMDLG32   (deferred)
00007ff8`54d10000 00007ff8`54dae000   sechost    (deferred)
00007ff8`54db0000 00007ff8`550dd000   combase    (deferred)
00007ff8`55140000 00007ff8`551f3000   KERNEL32   (pdb symbols)          c:\symbols\kernel32.pdb\07EFA0465AFB4214CBD5328B0B5ED0841\kernel32.pdb
00007ff8`55260000 00007ff8`55382000   RPCRT4     (deferred)
00007ff8`55390000 00007ff8`55433000   ADVAPI32   (deferred)
00007ff8`55440000 00007ff8`56939000   SHELL32    (deferred)
00007ff8`56970000 00007ff8`56b5d000   ntdll      (pdb symbols)          c:\symbols\ntdll.pdb\62C02F6EEF0AF78DFB36E22C396B9BBA1\ntdll.pdb
```

0:000>

Locals

Name	Value

Stack

Frame Index	Name
[0x0]	**notepad!WinMain**
[0x1]	notepad!__mainCRTStartup + 0x19f
[0x2]	KERNEL32!BaseThreadInitThunk + 0x14
[0x3]	ntdll!RtlUserThreadStart + 0x21

Locals | Watch | Threads | Stack | Breakpoints

Stack

- To see a

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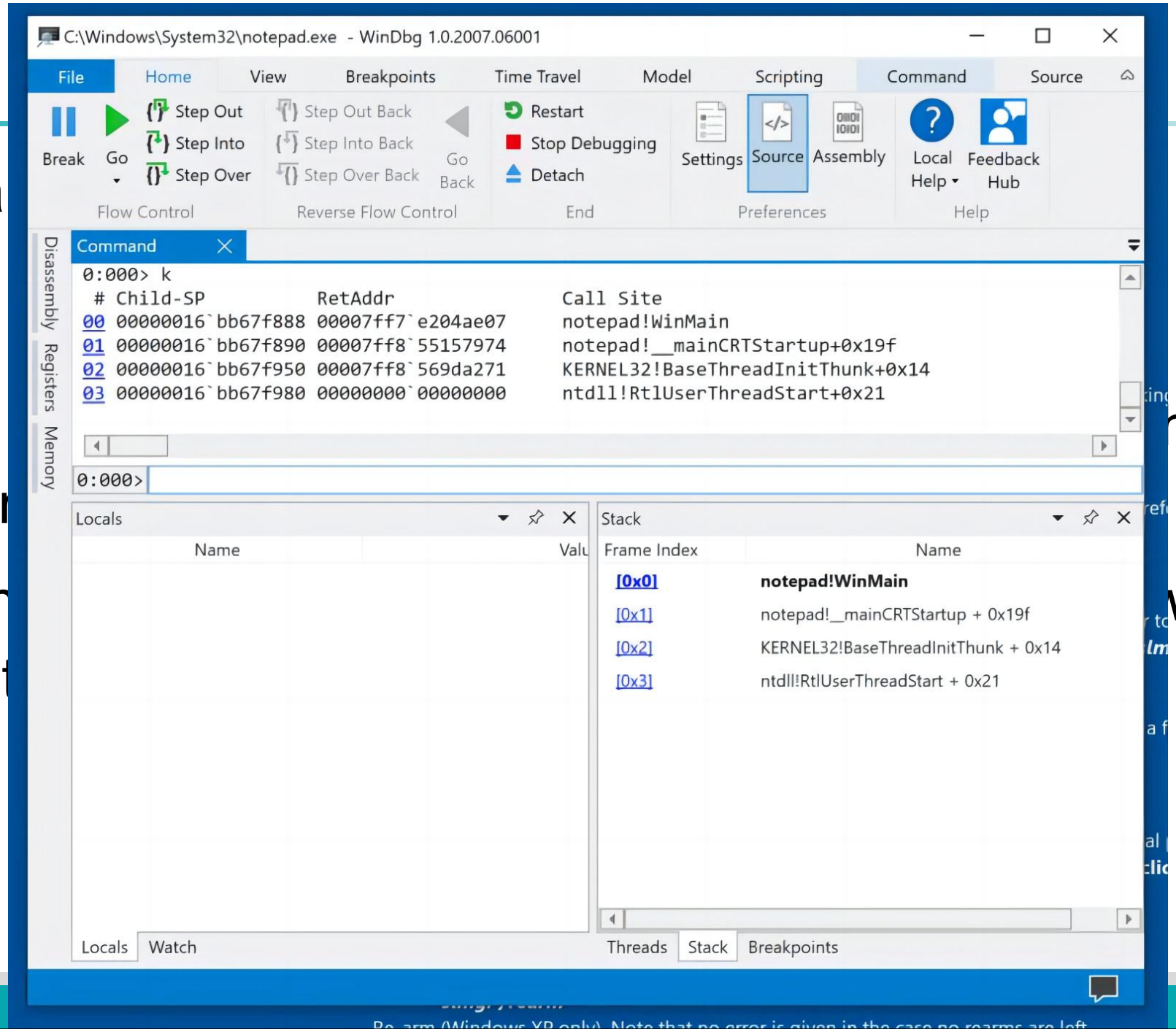
- You see his point, as shown

- This is th ver right pane, wit



Find

- Let's fi ll module

x ntdll

- You se s shown below.

C:\Windows\System32\notepad.exe - WinDbg 1.0.2007.06001

File | Home | View | Breakpoints | Time Travel | Model | Scripting | Command | Source

Break | Go | Step Out | Step Into | Step Over | Step Out Back | Step Into Back | Step Over Back | Go Back | Restart | Stop Debugging | Detach | Settings | Source | Assembly | Local Help | Feedback Hub

Flow Control | Reverse Flow Control | End | Preferences | Help

Disassembly | Registers | Memory

Command

```
ModLoad: 00007ff8`48b60000 00007ff8`48be9000   C:\Windows\System32\WINSPOOL.DRV
ModLoad: 00007ff8`52f10000 00007ff8`52f36000   C:\Windows\System32\bcrypt.dll
ModLoad: 00007ff8`4c1e0000 00007ff8`4c3b7000   C:\Windows\System32\urlmon.dll
ModLoad: 00007ff8`51ed0000 00007ff8`51f0d000   C:\Windows\System32\IPHLPAPI.DLL
ModLoad: 00007ff8`4bf30000 00007ff8`4c1d8000   C:\Windows\System32\iertutil.dll
ModLoad: 00007ff8`52380000 00007ff8`5238c000   C:\Windows\System32\CRYPTBASE.DLL
(1514.398): Break instruction exception - code 80000003 (first chance)
ntdll!LdrpDoDebuggerBreak+0x30:
00007ff8`56a42dfc cc              int     3
0:000> x ntdll!*CreateFile*
00007ff8`569dccbc ntdll!EtwpCreateFile (void)
00007ff8`56a101f0 ntdll!NtCreateFile (NtCreateFile)
00007ff8`56a101f0 ntdll!ZwCreateFile (ZwCreateFile)
00007ff8`56a90958 ntdll!_imp_NtCreateFile = <no type information>
```

0:000>

Locals

Name	Value

Stack

Frame Index	Name
[0x0]	**ntdll!LdrpDoDebuggerBreak + 0x30**
[0x1]	ntdll!LdrpInitializeProcess + 0x1d92
[0x2]	ntdll!_LdrpInitialize + 0x4e1dd
[0x3]	ntdll!LdrpInitialize + 0x3b
[0x4]	ntdll!LdrInitializeThunk + 0xe

Locals | Watch | Threads | Stack | Breakpoints

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File Edit Format View Help

C:\Windows\System32\notepad.exe - C:\Windows\System32\notepad.exe - WinDbg 1.0.2007.06001 (Administrator)

File Home View Breakpoints Time Travel Model Scripting Command Source

Break Go Step Out Step Into Step Over Step Out Back Step Into Back Step Over Back Go Back Restart Stop Debugging Detach Settings Source Assembly Local Help Feedback Hub

Flow Control Reverse Flow Control End Preferences Help

Disassembly Registers Memory

Command

```
     0 e Disable Clear  00007ff7`e2034238     0001 (0001)  2:**** notepad!WinMain
     1 e Disable Clear  00007ff8`56a101f0     0001 (0001)  2:**** ntdll!NtCreateFile
2:008> g
ModLoad: 00007ff8`53b90000 00007ff8`53bbe000   C:\Windows\System32\IMM32.DLL
Breakpoint 1 hit
ntdll!NtCreateFile:
00007ff8`56a101f0 4c8bd1          mov     r10,rcx
```

2:008>

Locals

Name	Value

Locals Watch

Stack

Frame Index	Name
[0x0]	**ntdll!NtCreateFile**
[0x1]	ntdll!LdrpMapResourceFile + 0x12b
[0x2]	ntdll!LdrMapAndVerifyResourceFile + 0x9a
[0x3]	ntdll!LdrLoadAlternateResourceModuleEx + 0x47c
[0x4]	ntdll!LdrpLoadResourceFromAlternativeModule + 0x...
[0x5]	ntdll!LdrpSearchResourceSection_U + 0x589

Threads Stack Breakpoints

Re-arm (all except Windows XP). Requires reboot.

slmgr /rearm

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C:\Windows\System32\notepad.exe - C:\Windows\System32\notepad.exe - WinDbg 1.0.2007.06001 (Administrator)

File | Home | View | Breakpoints | Time Travel | Model | Scripting | Command | Source

Break | Go | Step Out | Step Into | Step Over | Step Out Back | Step Into Back | Step Over Back | Go Back | Restart | Stop Debugging | Detach | Settings | Source | Assembly | Local Help | Feedback Hub

Flow Control | Reverse Flow Control | End | Preferences | Help

Disassembly | Registers | Memory

Command

```
. 8  Id: 1f6c.14b4 Suspend: 1 Teb: 00000049`d9858000 Unfrozen
2:008> k
 # Child-SP          RetAddr           Call Site
00 00000049`d97ad088 00007ff8`569b0d0b ntdll!NtCreateFile
01 00000049`d97ad090 00007ff8`569ade36 ntdll!LdrpMapResourceFile+0x12b
02 00000049`d97ad1b0 00007ff8`569a2a5c ntdll!LdrMapAndVerifyResourceFile+0x9a
03 00000049`d97ad230 00007ff8`569a22c9 ntdll!LdrLoadAlternateResourceModuleEx+0x47c
04 00000049`d97add40 00007ff8`569a3329 ntdll!LdrpLoadResourceFromAlternativeModule+0x1f1
2:008>
```

Locals

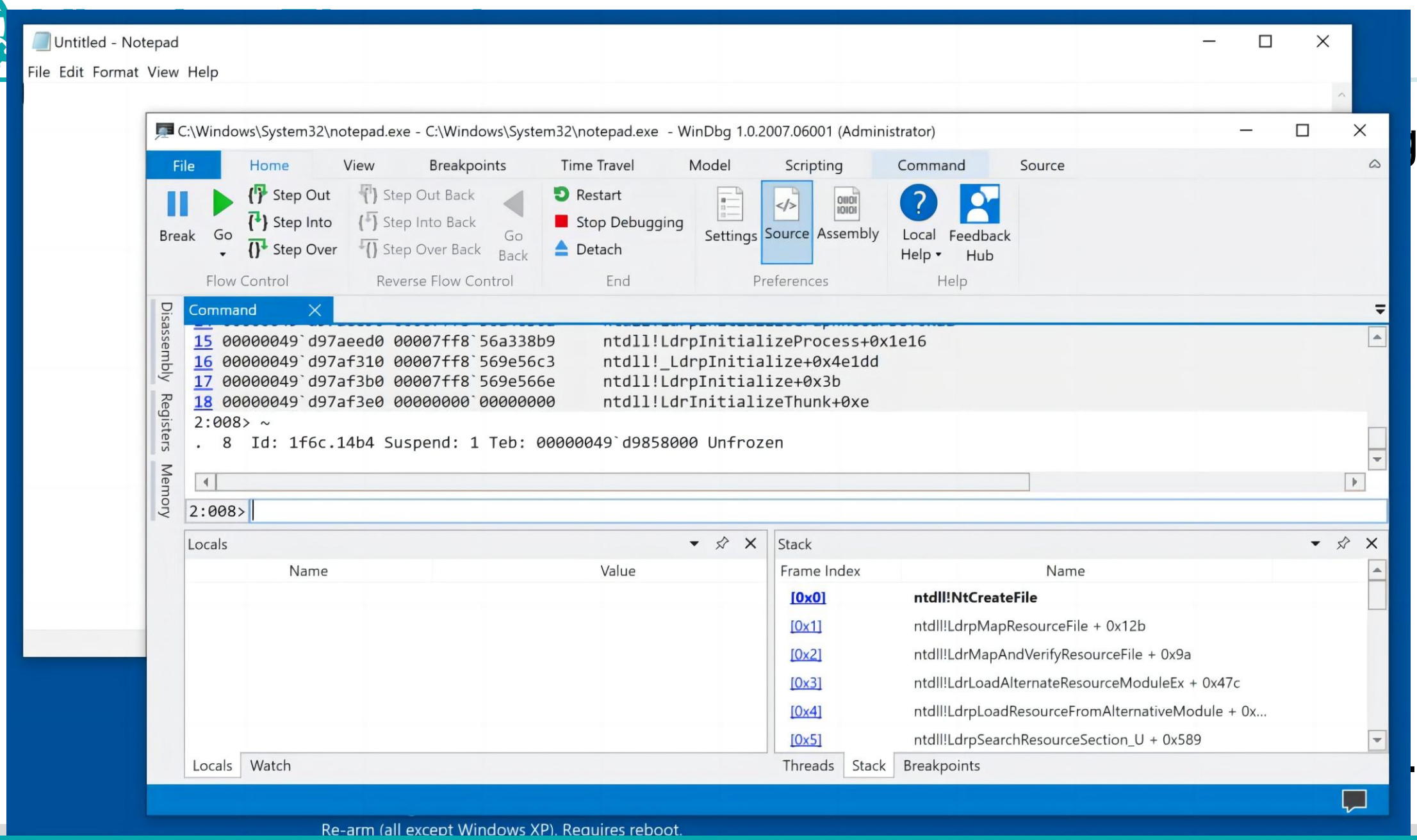
Name	Value

Stack

Frame Index	Name
[0x0]	**ntdll!NtCreateFile**
[0x1]	ntdll!LdrpMapResourceFile + 0x12b
[0x2]	ntdll!LdrMapAndVerifyResourceFile + 0x9a
[0x3]	ntdll!LdrLoadAlternateResourceModuleEx + 0x47c
[0x4]	ntdll!LdrpLoadResourceFromAlternativeModule + 0x...
[0x5]	ntdll!LdrpSearchResourceSection_U + 0x589

Locals | Watch | Threads | Stack | Breakpoints

Re-arm (all except Windows XP). Requires reboot.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File Edit Format View Help
C:\Windows\System32\notepad.exe - C:\Windows\System32\notepad.exe - WinDbg 1.0.2007.06001 (Administrator)
File Home View Breakpoints Time Travel Model Scripting Command Source
Break Go Step Out Step Into Step Over Step Out Back Step Into Back Step Over Back Go Back Restart Stop Debugging Detach Settings Source Assembly Local Help Feedback Hub
Flow Control Reverse Flow Control End Preferences Help
Disassembly Registers Memory
Command
15 00000049`d97aeed0 00007ff8`56a338b9 ntdll!LdrpInitializeProcess+0x1e16
16 00000049`d97af310 00007ff8`569e56c3 ntdll!_LdrpInitialize+0x4e1dd
17 00000049`d97af3b0 00007ff8`569e566e ntdll!LdrpInitialize+0x3b
18 00000049`d97af3e0 00000000`00000000 ntdll!LdrInitializeThunk+0xe
2:008> ~
. 8 Id: 1f6c.14b4 Suspend: 1 Teb: 00000049`d9858000 Unfrozen
2:008>
Locals
Name Value
Stack
Frame Index Name
[0x0] ntdll!NtCreateFile
[0x1] ntdll!LdrpMapResourceFile + 0x12b
[0x2] ntdll!LdrMapAndVerifyResourceFile + 0x9a
[0x3] ntdll!LdrLoadAlternateResourceModuleEx + 0x47c
[0x4] ntdll!LdrpLoadResourceFromAlternativeModule + 0x...
[0x5] ntdll!LdrpSearchResourceSection_U + 0x589
Locals Watch Threads Stack Breakpoints
Re-arm (all except Windows XP). Requires reboot.

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File Edit Format View Help

C:\Windows\System32\notepad.exe - C:\Windows\System32\notepad.exe - WinDbg 1.0.2007.06001 (Administrator)

File Home View Breakpoints Time Travel Model Scripting Command Source

Break Go Step Out Step Into Step Over Step Out Back Step Into Back Step Over Back Go Back Restart Stop Debugging Detach Settings Source Assembly Local Help Feedback Hub

Flow Control Reverse Flow Control End Preferences Help

Disassembly Registers Memory

Command

```
18 00000049`d97af3e0 00000000`00000000     ntdll!LdrInitializeThunk+0xe
2:008> ~
.  8  Id: 1f6c.14b4 Suspend: 1 Teb: 00000049`d9858000 Unfrozen
2:008> bl
     0 e Disable Clear  00007ff7`e2034238     0001 (0001)  2:**** notepad!WinMain
     1 e Disable Clear  00007ff8`56a101f0     0001 (0001)  2:**** ntdll!NtCreateFile
2:008>
```

Locals

Name	Value

Locals Watch

Stack

Frame Index	Name
[0x0]	**ntdll!NtCreateFile**
[0x1]	ntdll!LdrpMapResourceFile + 0x12b
[0x2]	ntdll!LdrMapAndVerifyResourceFile + 0x9a
[0x3]	ntdll!LdrLoadAlternateResourceModuleEx + 0x47c
[0x4]	ntdll!LdrpLoadResourceFromAlternativeModule + 0x...
[0x5]	ntdll!LdrpSearchResourceSection_U + 0x589

Threads Stack Breakpoints

Re-arm (all except Windows XP). Requires reboot.

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File Edit Format View Help

C:\Windows\System32\notepad.exe - C:\Windows\System32\notepad.exe - WinDbg 1.0.2007.06001 (Administrator)

File Home View Breakpoints Time Travel Model Scripting Command Source

Break Go Step Out Step Into Step Over Step Out Back Step Into Back Step Over Back Go Back Restart Stop Debugging Detach Settings Source Assembly Local Help Feedback Hub

Flow Control Reverse Flow Control End Preferences Help

Disassembly Registers Memory

Command

```
18 00000049`d97af3e0 00000000`00000000     ntdll!LdrInitializeThunk+0xe
2:008> ~
.  8  Id: 1f6c.14b4 Suspend: 1 Teb: 00000049`d9858000 Unfrozen
2:008> bl
     0 e Disable Clear  00007ff7`e2034238     0001 (0001)  2:**** notepad!WinMain
     1 e Disable Clear  00007ff8`56a101f0     0001 (0001)  2:**** ntdll!NtCreateFile
2:008>
```

Locals

Name	Value

Locals Watch

Stack

Frame Index	Name
[0x0]	**ntdll!NtCreateFile**
[0x1]	ntdll!LdrpMapResourceFile + 0x12b
[0x2]	ntdll!LdrMapAndVerifyResourceFile + 0x9a
[0x3]	ntdll!LdrLoadAlternateResourceModuleEx + 0x47c
[0x4]	ntdll!LdrpLoadResourceFromAlternativeModule + 0x...
[0x5]	ntdll!LdrpSearchResourceSection_U + 0x589

Threads Stack Breakpoints

Re-arm (all except Windows XP). Requires reboot.

THE END

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